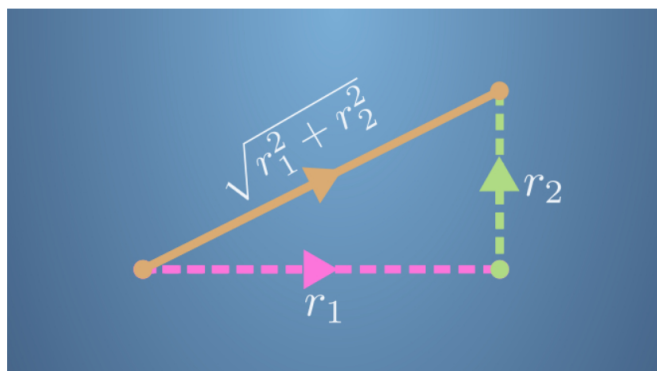


1. As we have seen in the lecture videos, the dot product of vectors has a lot of applications. Here, you will complete some exercises involving the dot product.

1 point

We have seen that the size of a vector with two components is calculated using Pythagoras' theorem, for example the following diagram shows how we calculate the size of the orange vector $\mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$:



In fact, this definition can be extended to any number of dimensions; the size of a vector is the square root

of the sum of the squares of its components. Using this information, what is the size of the vector $\mathbf{s} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 2 \end{bmatrix}$?

- ☐ $|\mathbf{s}| = \sqrt{10}$
- ☐ $|\mathbf{s}| = 30$
- ☐ $|\mathbf{s}| = 10$
- ☒ $|\mathbf{s}| = \sqrt{30}$

2. Remember the definition of the dot product from the videos. For two n component vectors, $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_nb_n$.

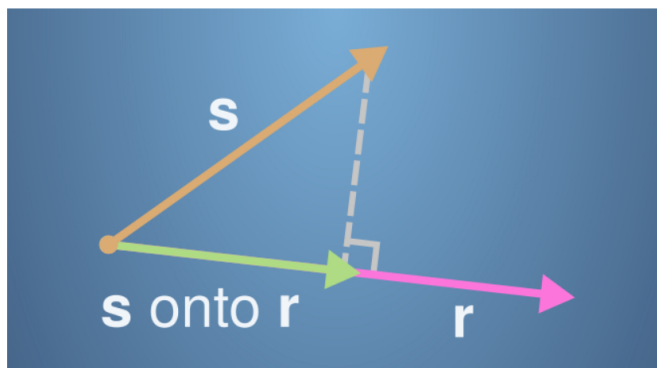
1 point

What is the dot product of the vectors $\mathbf{r} = \begin{bmatrix} -5 \\ 3 \\ 2 \\ 8 \end{bmatrix}$ and $\mathbf{s} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}$?

- ☐ $\mathbf{r} \cdot \mathbf{s} = \begin{bmatrix} -5 \\ 6 \\ -2 \\ 0 \end{bmatrix}$
- ☐ $\mathbf{r} \cdot \mathbf{s} = \begin{bmatrix} -4 \\ 5 \\ 1 \\ 9 \end{bmatrix}$
- ☐ $\mathbf{r} \cdot \mathbf{s} = 1$
- ☒ $\mathbf{r} \cdot \mathbf{s} = -1$

3. The lectures introduced the idea of projecting one vector onto another. The following diagram shows the projection of $\mathbf{s}$ onto $\mathbf{r}$ when the vectors are in two dimensions:

1 point



Remember that the scalar projection is the size of the green vector. If the angle between $\mathbf{s}$ and $\mathbf{r}$ is greater than $\pi/2$, the projection will also have a minus sign.

we can do projection in any number of dimensions. Consider two vectors with three components,

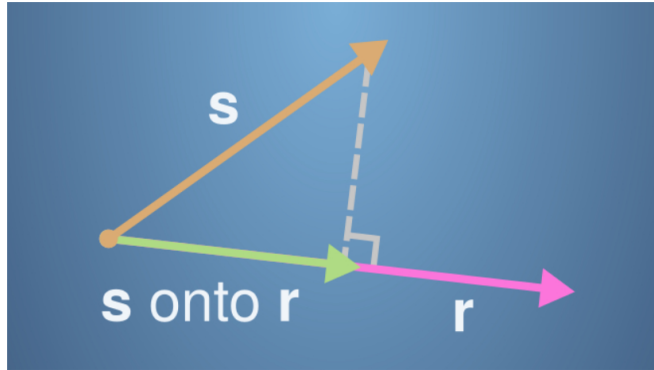
$$\mathbf{r} = \begin{bmatrix} 3 \\ -4 \\ 0 \end{bmatrix} \text{ and } \mathbf{s} = \begin{bmatrix} 10 \\ 5 \\ -6 \end{bmatrix}.$$

What is the scalar projection of $\mathbf{s}$ onto $\mathbf{r}$?

- ☐ $\frac{1}{2}$
- ☐ $-\frac{1}{2}$
- ☐ -2
- ☒ 2

4. Remember that in the projection diagram, the vector projection *is* the green vector:

1 point



$$\text{Let } \mathbf{r} = \begin{bmatrix} 3 \\ -4 \\ 0 \end{bmatrix} \text{ and let } \mathbf{s} = \begin{bmatrix} 10 \\ 5 \\ -6 \end{bmatrix}.$$

What is the vector projection of $\mathbf{s}$ onto $\mathbf{r}$?

- ☐ $\begin{bmatrix} 6 \\ -8 \\ 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 6 \\ 4 \\ 0 \end{bmatrix}$
- ☒ $\begin{bmatrix} 6/5 \\ -8/5 \\ 0 \end{bmatrix}$
- ☐ $\begin{bmatrix} 30 \\ -20 \\ 0 \end{bmatrix}$

5. Let $\mathbf{a} = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 5 \\ 12 \end{bmatrix}$.

1 point

Which is larger, $|\mathbf{a} + \mathbf{b}|$ or $|\mathbf{a}| + |\mathbf{b}|$?

- ☒ $|\mathbf{a} + \mathbf{b}| < |\mathbf{a}| + |\mathbf{b}|$
- ☐ $|\mathbf{a} + \mathbf{b}| > |\mathbf{a}| + |\mathbf{b}|$
- ☐ $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$

6. Which of the following statements about dot products are correct?

1 point

- ☐ The scalar projection of $\mathbf{s}$ onto $\mathbf{r}$ is always the same as the scalar projection of $\mathbf{r}$ onto $\mathbf{s}$.
- ☐ The order of vectors in the dot product is important, so that $\mathbf{s} \cdot \mathbf{r} \neq \mathbf{r} \cdot \mathbf{s}$.
- ☒ We can find the angle between two vectors using the dot product.
- ☒ The vector projection of $\mathbf{s}$ onto $\mathbf{r}$ is equal to the scalar projection of $\mathbf{s}$ onto $\mathbf{r}$ multiplied by a vector of unit length that points in the same direction as $\mathbf{r}$.
- ☒ The size of a vector is equal to the square root of the dot product of the vector with itself.